

Answer Key — Natural Resources

Final answers with a one-line reason. Full method is in the Notes.

Objective

- Q1 (c) **Iron ore** — a metal ore, formed over millions of years; non-renewable.
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- Q2 (c) **Total available resources** — stock = whole reserve, in use or not.
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- Q3 (a) **1-B, 2-A** — Patent = right over invention; Sikkim = first organic state.
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- Q4 **Utility** — usefulness is what makes something a resource.
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- Q5 **Patent** — exclusive right over an idea/invention.
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- Q6 **Sikkim** — became 100% organic in 2016.
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- Q7 (a) Both correct and R explains A — over-use beyond recovery rate depletes the stock.
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- Q8 (b) Abundant resources don't guarantee prosperity — the "paradox of plenty".
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- Q9 (b) **culturally acceptable** — the third of the three tests.
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- Q10 **France** — India & France launched the ISA in 2015.
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Short Answer

- Q11 An element of Nature becomes a resource when humans use it. Three conditions: **technologically accessible, economically feasible, culturally acceptable**.
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- Q12 **Renewable** = restored/regenerated within our timescale if not overused (e.g. solar/wind/timber).
Non-renewable = formed over millions of years, can't be replaced at our rate of use (e.g. coal/petroleum/iron).
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- Q13 Ecosystem services = benefits humans *receive* from Nature's functions — e.g. clean water from a forest that filters it (also pollinated crops, protected farmland).
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Long Answer

- Q14 Meet present needs without harming future generations. Principles: (1) restore & regenerate renewables, don't exceed recovery rate; (2) use non-renewables judiciously till greener options arrive; (3) ensure fair access to basic resources for all.
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Q15 Resource curse / "paradox of plenty" = resource-rich regions can stay poor because they can't convert resources into higher-value products. India avoided it by **investing in industries, technology and skills**.

Q16 Functions: producing oxygen, filtering water, preventing soil erosion, providing habitat, pollinating crops (any 3). Uneven distribution → competition for scarce resources → displacement, threatened sacred sites, water-sharing disputes → conflict/war.

Very Long Answer

Q17 **Punjab:** Green Revolution's HYV wheat/paddy needed more water; free power → over-pumping of groundwater; chemical fertilisers/pesticides polluted it. Result: ~80% of Punjab "over-exploited", water below ~30 m, health hazards. Lesson: short-term food security caused long-term unsustainability — use within recovery limits.

OR — Cement: cement is vital but highly polluting (dust harms lungs, leaves, soil, water). CPCB issues guidelines. Sustainable move: revive traditional stone & mud, add plant-based and recycled materials, combine traditional methods with modern technology → less pollution, local jobs, climate-suited buildings.

Passage

Q18 (i) Nature ignores political boundaries, so states/countries compete for the same water. (ii) Karnataka, Tamil Nadu, Kerala, Puducherry. (iii) Negotiations and deft/careful management (fair sharing agreements).
